Which dashboard do I need?

A beginner's map for verifying NBA Criteria 3, 4, 5 and 6

You do not need to know anything about NBA accreditation or about this code. For every criterion this guide tells you **which page to open, which person to sign in as, and what to click**.

If you want to…	Go to
Understand the pages and who sees what	Section 1 (page 2)
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Status of this guide
Criterion 3 is built and tested. Criteria 4, 5 and 6 are still being built in a separate work stream. This guide names the dashboards and tabs that exist today. If a tab is renamed later, the answer to *which dashboard* stays the same. The guide is produced by `docs/tools/build_dashboard_guide.py` and can be regenerated after those criteria are finished.

1. The pages and who can open them

The system has **three kinds of people** (roles) and **eleven pages** you will use. A **criterion** is one chapter of the NBA checklist (Criterion 3 is about learning outcomes, 4 about students, 5 about faculty, 6 about faculty contributions). Each chapter is a list of **requirements**, and each requirement ends as **PASS**, **WARNING** or **FAIL**.

Page	Address after the site opens	Who	What it is for
Admin dashboard	/dashboards/admin.html	Admin	Approve users, assign HOD, backups, audit log, whole-institution overview
HOD dashboard	/dashboards/hod.html	HOD, Admin	Department health, approvals, course assignments, gaps and actions
Faculty dashboard	/dashboards/faculty.html	Faculty, HOD, Admin	Your own subjects, marks, profile, research records and evidence uploads
Criterion 3 page	/dashboards/c3.html	Everyone signed in; HOD and Admin can evaluate	Where Criterion 3 is verified
Criterion 4 page	/dashboards/c4.html	same	Where Criterion 4 is verified
Criterion 5 page	/dashboards/c5.html	same	Where Criterion 5 is verified
Criterion 6 page	/dashboards/c6.html	same	Where Criterion 6 is verified
Virtual Inspection	/dashboards/inspection.html	HOD, Admin	Runs every requirement of Criteria 3 to 6 and gives one readiness score
Reports	/dashboards/reports.html	Everyone signed in; export for HOD and Admin	Download PDF or Excel report
Login, Register, Pending	login.html, register.html, pending.html	Anyone	Sign in, ask for an account, wait for approval

The one idea that makes everything simple

Verifying a criterion always happens on its own **Criterion page** (c3, c4, c5 or c6), signed in as HOD or Admin. The Faculty, HOD and Admin dashboards are where the **data comes from** and where people are managed. If the sample data is loaded, the Criterion page alone is enough to verify.

2. Start the system and sign in

Start

Windows (PowerShell, in the project folder): `.\start.ps1 setup` the first time, then `.\start.ps1`.
Mac or Linux: `./start.sh setup` the first time, then `./start.sh`.

Open `http://127.0.0.1:5500` in your browser. Type exactly **127.0.0.1**; the word *localhost* can make sign-in fail with "Failed to fetch".

Sample logins (password for all is `password123`)

Sign in as	Email	Use it for
Admin	admin@college.edu	Everything, plus users, years and backups
HOD	hod@college.edu	Verifying criteria, approvals, gaps, inspection
Faculty	faculty1@college.edu … faculty6@college.edu	Entering marks, profile, research records, evidence

The login page also has demo buttons that fill these in for you. The first start loads a sample department (Computer Science and Engineering, 2026-27) with 6 faculty, 60 students and 5 subjects. It is deliberately imperfect so you can see the system find problems. Change the passwords before any real use.

Words you will see

Word	Plain meaning
NBA	The body that accredits engineering programmes. This system checks how ready a department is.
Criterion	One chapter of the NBA checklist. This guide covers Criteria 3, 4, 5 and 6.
Requirement	One line in a criterion, for example 5.1 Student-faculty ratio. It ends as PASS, WARNING or FAIL.
Evaluate	A button that reads the stored records, applies the NBA rule and shows marks and a reason.
Inputs	A form for facts the system cannot work out alone (for example a reviewer score).
Evidence	A document (certificate, result sheet, report) that proves a requirement.
Gap	A requirement that is failing or has no evidence. HOD turns gaps into assigned actions.
Role	Admin, HOD or Faculty. Your role decides which buttons you see.

3. The master table: which dashboards for which criterion

Read across a row to see everything one criterion touches. **REQUIRED** means you cannot verify without it. **ENTER DATA** means the numbers are typed there, so open it only if you want to change results. **Optional** pages add context.

Criterion	Its Criterion page	Faculty dashboard	HOD dashboard	Admin dashboard	Virtual Inspection
Criterion 3 Outcome-Based Assessment	**REQUIRED**	**ENTER DATA**	Optional	One-time setup	Optional
Criterion 4 Students' Performance	**REQUIRED**	**ENTER DATA**	Optional	Not needed	Optional
Criterion 5 Faculty Information	**REQUIRED**	**ENTER DATA**	Optional	One-time setup	Optional
Criterion 6 Faculty Contributions	**REQUIRED**	**ENTER DATA**	Optional	Not needed	Optional

Shortest path for each criterion (verification only)

- **Criterion 3:** HOD or Admin, then the **Criterion 3 page**.
- **Criterion 4:** HOD or Admin, then the **Criterion 4 page**.
- **Criterion 5:** HOD or Admin, then the **Criterion 5 page**.
- **Criterion 6:** HOD or Admin, then the **Criterion 6 page**.
- **All four at once:** HOD or Admin, then the **Virtual Inspection** page and press **Run Virtual NBA Inspection**.

Why not just the Admin dashboard?

The Admin dashboard manages *people and the system* (approvals, roles, backups, audit log). Its Institution Overview shows a summary of the four criteria, but the proof for a single requirement (the marks, the reason, the inputs, the history) lives on that criterion's own page. Signing in as Admin gives you access to all of those pages, so Admin is a fine login for verification; the Admin *dashboard* is just not where the verifying happens.

Who can click what on a Criterion page

Action on a Criterion page	Faculty	HOD	Admin
Look at the requirement table, **History**, **View inputs**	Yes	Yes	Yes
Evaluate a requirement, **Evaluate all**	No	Yes	Yes
Type and save **Inputs**	No	Yes	Yes
Add own records (marks, qualification, publication, certificate)	Yes	Yes	Yes
Approve a method (for example C3 attainment) or link evidence to a requirement	No	Yes	Yes

4. Criterion 3: Outcome-Based Assessment

120 marks

Status: Built and tested

The question it answers: Did students actually learn what we said they would learn?

> **Minimum to verify this criterion: Criterion 3 page**
> Sign in as HOD or Admin. Open **Criterion 3 (Assessment)** from the left menu. With the sample data loaded, no other dashboard is needed to verify it.

Dashboards for this criterion

Dashboard	Sign in as	Where exactly and why	Needed?
Criterion 3 page	HOD or Admin	Left menu: **Criterion 3 (Assessment)**. Tabs: Requirements & Evaluation, CO Attainment & Matrix, Marks & Attendance, Evidence Documents, AI Advisory & RAG.	**REQUIRED**
Faculty dashboard	Faculty	Tab **My Assigned Courses (C3)**: add course outcomes, enter marks. Only needed if you want to change the numbers.	**ENTER DATA**
HOD dashboard	HOD	Tab **Course Assignments (C3)**: decide which faculty teaches which subject.	Optional
Admin dashboard	Admin	Only to approve new users or create the academic year before data entry.	One-time setup
Virtual Inspection	HOD or Admin	Shows Criterion 3 next to 4, 5 and 6 with an overall score.	Optional

Verify it, step by step

1	Sign in as **HOD** (`hod@college.edu`).
2	Left menu, **Criterion 3 (Assessment)**. You land on **Requirements & Evaluation**.
3	On row **3.1** click **Evaluate**. Read the status, the marks and the reason.
4	Rows 3.2 to 3.6 say they need a reviewer score. Click **Inputs**, type a score, **Save**. The row re-evaluates by itself.
5	Open tab **CO Attainment & Matrix**. Weak subjects are flagged. Click **Approve attainment method**; rows 3.7 and 3.8 now get a real score.
6	Click **History** on any row to see every earlier evaluation. Nothing is overwritten.

What good looks like

Rows 3.1 to 3.8 show PASS or WARNING with marks. The sample data has two weak subjects (CS503 and CS504), so a few WARNING or FAIL rows are expected and are the proof that the rules are real.

Prove it is real

Sign in as Faculty, raise the marks of CS503 students above 60 percent, then re-open **CO Attainment & Matrix**. The weak flag clears and the score rises.

Requirements in this criterion

3.1 Continuous assessment · 3.2 Exam papers · 3.3 Lab work · 3.4 Industrial training · 3.5 Projects · 3.6 SDG mapping · 3.7 CO attainment · 3.8 PO and PSO attainment

5. Criterion 4: Students' Performance

120 marks

Status: Being built in a separate work stream

The question it answers: How are our students performing?

> **Minimum to verify this criterion: Criterion 4 page**
> Sign in as HOD or Admin. Open **Criterion 4 (Students)**. Marks and attendance come from Faculty entries or the sample data.

Dashboards for this criterion

Dashboard	Sign in as	Where exactly and why	Needed?
Criterion 4 page	HOD or Admin	Left menu: **Criterion 4 (Students)**. Tabs: Requirements (4.1 to 4.7), Success Rate & Academic Index, Placements & Higher Studies, Evidence Checklist & Result Sheets, AI Advisory & Gap Analysis.	**REQUIRED**
Faculty dashboard	Faculty	Tab **Assigned Students (C4)**: enter attendance and see students who need support.	**ENTER DATA**
HOD dashboard	HOD	Tab **Students Performance (C4)**: pass rates, marks, at-risk students (a read-only check).	Optional
Admin dashboard	Admin	Not needed for verification.	Not needed
Virtual Inspection	HOD or Admin	Confirms Criterion 4 inside the whole-department score.	Optional

Verify it, step by step

1	Sign in as **HOD**.
2	Left menu, **Criterion 4 (Students)**. Click **Evaluate** on row 4.3 (academic performance). The marks come from the average student marks.
3	Rows 4.1 and 4.2 ask for lists of numbers. Click **Inputs**, fill the listed fields, **Save**.
4	Row 4.6 (placement) says **Needs rule clarification**. That is correct: the NBA manual leaves the formula to the institution.
5	Open the **Students** area to see the at-risk list: attendance under 75 percent or marks under 40 percent, with the reason shown.
6	To see the rule react: sign in as Faculty, lower a few attendance values, and return to the HOD view.

What good looks like

4.3 to 4.5 show marks calculated from student data. The at-risk list names each student and the reason. 4.6 honestly says it needs a decision.

Prove it is real

Lower some attendance below 75 percent as Faculty. The at-risk list grows.

Requirements in this criterion

4.1 Enrolment ratio · 4.2 Success rate without backlogs · 4.3 / 4.4 / 4.5 Academic performance (1st, 2nd, 3rd year) · 4.6 Placement and higher studies · 4.7 Professional activities

6. Criterion 5: Faculty Information

100 marks

Status: Being built in a separate work stream

The question it answers: Do we have enough qualified faculty?

> **Minimum to verify this criterion: Criterion 5 page**
> Sign in as HOD or Admin. Open **Criterion 5 (Faculty)**. Almost everything is calculated from faculty profiles, so nothing else is needed.

Dashboards for this criterion

Dashboard	Sign in as	Where exactly and why	Needed?
Criterion 5 page	HOD or Admin	Left menu: **Criterion 5 (Faculty)**. Tabs: Requirements (5.1 to 5.5), Faculty Roster & Profile Rings, SFR, FQI & Cadre Gauges, Workload & Assignments.	**REQUIRED**
Faculty dashboard	Faculty	Tab **My C5 Profile & FQI**: keep qualifications, designation and experience complete.	**ENTER DATA**
HOD dashboard	HOD	Tab **Faculty Strength & SFR (C5)**: faculty count and student-faculty ratio at a glance.	Optional
Admin dashboard	Admin	Tab **Pending Approvals**: a new faculty member only counts after approval, which changes the ratio.	One-time setup
Virtual Inspection	HOD or Admin	Confirms Criterion 5 inside the whole-department score.	Optional

Verify it, step by step

1	Sign in as **HOD**.
2	Left menu, **Criterion 5 (Faculty)**. Click **Evaluate** on 5.1. With 60 students and 6 faculty the ratio is 10 to 1, which earns full marks.
3	Evaluate **5.2** (qualification index) and **5.3** (cadre proportion). Each shows the numbers it used.
4	Row 5.4 asks for two inputs (provision exists, yearly hours). Use **Inputs**.
5	Row 5.5 (retention) says **Needs rule clarification**. Expected.
6	To see the rule react: as Faculty change a designation to Assistant Professor, then Evaluate 5.3 again. The cadre score drops.

What good looks like

5.1 to 5.3 show a number and a reason on the first click, without any typing. That proves they are computed, not typed in.

Prove it is real

Sign in as Admin, approve one more faculty account, then Evaluate 5.1 and 5.2 again. The ratio and the index change.

Requirements in this criterion

5.1 Student-faculty ratio · 5.2 Faculty qualification index · 5.3 Faculty cadre proportion · 5.4 Visiting / adjunct faculty · 5.5 Faculty retention

7. Criterion 6: Faculty Contributions

120 marks

Status: Being built in a separate work stream

The question it answers: What are our faculty contributing (research, patents, projects, consultancy)?

Minimum to verify this criterion: Criterion 6 page
Sign in as HOD or Admin. Open **Criterion 6 (R&D)**. Publications, patents and projects come from Faculty entries or the sample data.

Dashboards for this criterion

Dashboard	Sign in as	Where exactly and why	Needed?
Criterion 6 page	HOD or Admin	Left menu: **Criterion 6 (R&D)**. Tabs: Requirements (6.1 and 6.2), Year-wise R&D Stacked Chart, Faculty Leaderboard, Certificate Versioning.	**REQUIRED**
Faculty dashboard	Faculty	Tab **My C6 Research & Consultancy**: add each publication, patent, project and consultancy record. Also **Evidence Upload & Lifecycle** for certificates.	**ENTER DATA**
HOD dashboard	HOD	Tab **Research & Contributions (C6)**: department totals.	Optional
Admin dashboard	Admin	Not needed for verification.	Not needed
Virtual Inspection	HOD or Admin	Confirms Criterion 6 inside the whole-department score.	Optional

Verify it, step by step

1	Sign in as **HOD**.
2	Left menu, **Criterion 6 (R&D)**. Click **Evaluate** on 6.2.1 (publications), 6.2.3 (patents), 6.2.4 (sponsored research) and 6.2.5 (consultancy). These are calculated from faculty records.
3	Rows such as 6.1.2.2 (FDP participation) ask for numbers. Use **Inputs**.
4	Open **Year-wise R&D Stacked Chart** and **Faculty Leaderboard** to see the same records as pictures.
5	To see the rule react: sign in as **faculty1**, open **My C6 Research & Consultancy**, add a publication. Back as HOD, Evaluate 6.2.1 again; the count goes up.
6	Upload an FDP certificate as Faculty (**Evidence Upload & Lifecycle**). As HOD, link it to row 6.1.2.2 using the evidence option on that row.

What good looks like

6.2.x rows show counts and marks straight from the stored records. Linking a certificate changes the row to **Has evidence**.

Prove it is real

Add a patent with status Granted as Faculty. Evaluate 6.2.3 as HOD: marks increase.

Requirements in this criterion

6.1.1 Societies · 6.1.2.x Resource persons and FDPs · 6.1.3 MOOCs developed · 6.1.4 MOOC certification · 6.1.5 FDPs organised · 6.1.6 Student innovation · 6.1.7 Industry training · 6.2.1 Publications · 6.2.2 PhD guidance · 6.2.3 Patents · 6.2.4 Sponsored research · 6.2.5 Consultancy · 6.2.6 Seed money

8. A 15-minute check of all four criteria

Tick each box as you go. Everything here is done as **HOD** unless it says otherwise.

[ ]	Open `http://127.0.0.1:5500`, sign in as **hod@college.edu**.
[ ]	Open **Virtual Inspection** (button on the HOD dashboard, tab *Department Analytics & SAR*, or the drop-down at the top). Press **Run Virtual NBA Inspection**. Note the readiness percentage.
[ ]	**Criterion 3:** Evaluate 3.1; set an input on 3.2; open *CO Attainment & Matrix* and approve the method.
[ ]	**Criterion 4:** Evaluate 4.3; read the at-risk students list.
[ ]	**Criterion 5:** Evaluate 5.1, 5.2, 5.3 and read the reasons.
[ ]	**Criterion 6:** Evaluate 6.2.1, 6.2.3, 6.2.4, 6.2.5.
[ ]	Sign out. Sign in as **faculty1@college.edu**. Add one publication under *My C6 Research & Consultancy*.
[ ]	Sign in as HOD again. Evaluate 6.2.1. The count should be one higher.
[ ]	Open **Gaps & Actions** on the HOD dashboard. Press **Generate gaps**. Assign one action with an owner and a due date.
[ ]	Run the **Virtual Inspection** again. The score should be higher than in step 2.
[ ]	Open **Reports** and download the PDF.

What the status words mean

Status	Meaning	What you do
PASS	Meets the threshold	Keep evidence current
WARNING	Below full marks, or an input is missing	Read the reason; enter the missing input
FAIL	Clearly below the expected level	Generate gaps and assign a corrective action
Needs rule clarification	The NBA manual leaves a choice to the institution	Approve a method or record the decision
Not applicable	Does not apply this year	Nothing

On an empty department the Inspection says **NOT READY, 0 percent**. That is the honest answer. Readiness labels: **READY** at 85 percent or more with no failures, **NEARLY READY** 60 to 84 percent, **NOT READY** below 60.

9. If something does not work

What you see	What to do
Login says "Failed to fetch", or pages stay empty	Open the site as `127.0.0.1:5500`, not localhost. Check the backend: `http://127.0.0.1:8000/api/v1/health` should answer.
A new faculty account cannot sign in	An Admin or HOD must approve it first (Admin dashboard, Pending Approvals).
"Missing context keys" after Evaluate	Click **Inputs** on that row, fill the fields listed, **Save**.
3.7 and 3.8 say Needs rule clarification	On Criterion 3, tab *CO Attainment & Matrix*, press **Approve attainment method**.
No **Evaluate** button on the Criterion page	You are signed in as Faculty. Sign in as HOD or Admin.
A page says you are not allowed	That page needs a higher role. Admin dashboard needs Admin.
Numbers look round and never change	That tab may still show sample values. Trust the Evaluate result on the Criterion page.
A tab named in this guide is missing	Criteria 4, 5 and 6 are being built; the tab may have been renamed. Use the same Criterion page and look for the closest name.
Upload is rejected	Use PDF, DOCX, XLSX, CSV, TXT, PNG or JPEG, under 25 MB.

Where to read more

docs/USER_GUIDE.md (full manual, also available as a PDF) explains every requirement and how marks are calculated. **docs/CRITERION_3_QUICK_TEST.md** is a short test script for Criterion 3 with sample evidence files.